

Load Cell Transducer

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Introduction:

Load cell is a transducer which converts tension or compression into an electric signal. load cells can be classified according to their operating principle into three main types: strain gauge load cells, hydraulic load cells, and pneumatic load cell.

-Strain Gauge Load Cell:

The most common type. It converts mechanical deformation (which changes its Resistance slightly) into an electrical signal using strain gauges arranged in a Wheatstone bridge which can be shaped with **N** strain gauges (half, quarter or full bridge).

-Hydraulic Load Cell:

Uses a fluid-filled chamber. When force is applied, pressure increases in the fluid, and this pressure change is measured to determine the load.

-Pneumatic Load Cell:

Works with air pressure instead of liquid. When a load is applied, air pressure changes proportionally, and the pressure is converted into a readable output.

We will focus on strain gauge load cell because they give consistent readings over long periods, even in changing environmental conditions. Also, They are cheaper to produce and maintain than hydraulic or pneumatic load cells. But it is the only which needs an external excitation V_i .

When a load or force is applied, the strain gauge undergoes a slight change in resistance. These strain gauges are connected in a Wheatstone Bridge configuration, producing a small differential voltage output that is proportional to the applied force. It should be noted that V_{out} is zero when no deformation ($\Delta R = 0$). The strain gauges $\Delta R/R$ ratio is small ($\Delta R \ll R$), so the output voltage will consequently be very small ($V_o = (N/4) \cdot V_i \cdot (\Delta R/R)$ will be proved). So, this signal must be amplified and filtered before being measured by an Analog-to-Digital Converter or a data acquisition system (such as our Analog Discovery).

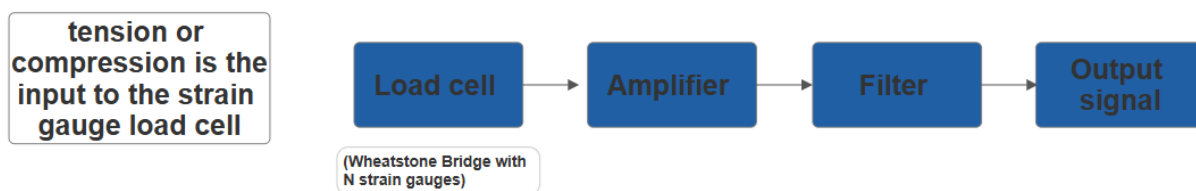
Literature Survey:

Recent studies focus on improving the performance of amplification and filtering stages to increase measurement precision and reduce noise. Differential and instrumentation amplifiers are commonly used to provide accurate signal conditioning, while active low-pass filters help to remove high-frequency interference. These developments make strain gauge load cells practical for modern digital systems such as microcontrollers and data acquisition (DAQ) devices. In conclusion, the reviewed literature shows that strain gauge load cells, when combined with efficient amplification and filtering circuits, remain an accurate and economical choice for measuring mechanical force.

Applications of Load Cells:

Load cells are used in many areas to measure force or weight accurately. They are found in digital weighing machines, industrial process control, and robotics to monitor motion and gripping force. Also, In the automotive and aerospace fields, they help test material strength and structural loads. They are also used in medical devices and laboratory experiments for precise force measurement.

Block diagram for the strain gauge load cell:

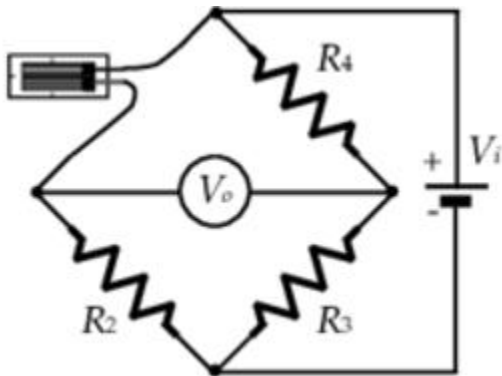


The system starts with a load cell which converts the applied mechanical force into a small electrical signal. This signal is then sent to an amplifier to increase its strength. After amplification, the signal goes to a filter to remove any noise and make it stable. Eventually, the clean and conditioned signal is sent to an Analog-to-Digital Converter or a Data Acquisition device for measurement and any desired analysis.

Theoretical Background:

Wheatstone bridge could be quarter (1 strain gauge) , half (2 strain gauges), and full (4 strain gauges).

Quarter bridge (1 strain gauge) :



$$R_2 = R_3 = R_4 = R$$

$R_1 = R$ (strain gauge) , which changes due to external acting force

$$\text{So, } R_1 = R + \Delta R$$

Using voltage divider we can get that : $V_o(\text{output}) = \frac{V_i}{2} - V_i \frac{R}{2R + \Delta R}$

$$\text{So, } V_o = \frac{V_i}{2} \frac{1}{\frac{2R}{\Delta R} + 1} \quad \text{we said that the change is slight } \Delta R \ll R \text{ so, } \frac{2R}{\Delta R} \text{ is}$$

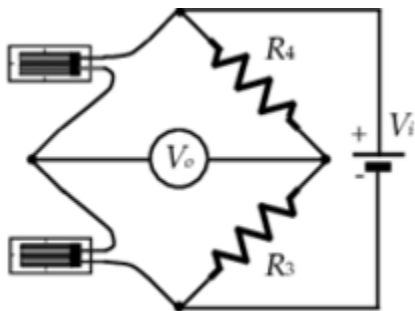
$$\text{big relative to 1, and 1 can be ignored. So, } V_o \approx \frac{V_i}{4} \frac{\Delta R}{R}$$

We can see from this equation that the output voltage is proportional to the change in strain gauge resistance, also proportional to the excitation voltage V_i .

The change in resistance (ΔR) of a strain gauge depends on the change in its length (ΔL). The strain is given by $\frac{\Delta L}{L}$, and for elastic material, stress is proportional to strain. As the applied force increases, both stress and strain increase, causing a proportional change in resistance. Thus, ΔR is directly proportional to the applied force. So, we can conclude that :

$$V_o(\text{output}) \propto \Delta R \propto \Delta L \propto F$$

half bridge (2 strain gauge) :



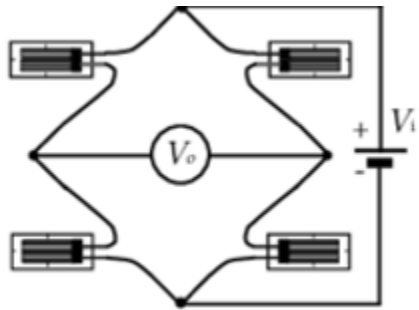
$R_3=R_4=R$, one of the strain gauges will get longer and the other will get shorter, meaning that : $R_1=R+\Delta R$, $R_2=R-\Delta R$

Using voltage divider we can get that : $V_o(\text{output})=V_i \frac{R+\Delta R}{2R} - \frac{V_i}{2}$

$$\text{So, } V_o = \frac{V_i}{2} \frac{\Delta R}{R}$$

Hence, we can notice that the half Wheatstone bridge is more sensitive than the quarter one (V_o is doubled)

Full bridge (4 strain gauge) :



Two of the strain gauges will get longer and the others will get shorter, meaning that :

$$R_1=R+\Delta R, R_2=R-\Delta R, R_3=R+\Delta R, R_4=R-\Delta R$$

Using voltage divider we can get that : $V_o(\text{output})=V_i \frac{R+\Delta R}{2R} - V_i \frac{R-\Delta R}{2R}$

$$\text{So, } V_o = V_i \frac{\Delta R}{R}$$

From the three cases we can conclude that :

$$V_o = \frac{N}{4} V_i \frac{\Delta R}{R}, \text{ Where } N \text{ is number of strain gauges.}$$

Hence, the more N used, the more sensitive the load cell is.

So, it is better to use a Full Wheatstone bridge than others.

Also, 4 strain gauges will cancel temperature effects

because the whole system will be affected equally

(assuming that the 4 strain gauges are of the same material)

-After this, the output V_o will be amplified and filtered to get a clean signal with no noise, and can be detected.